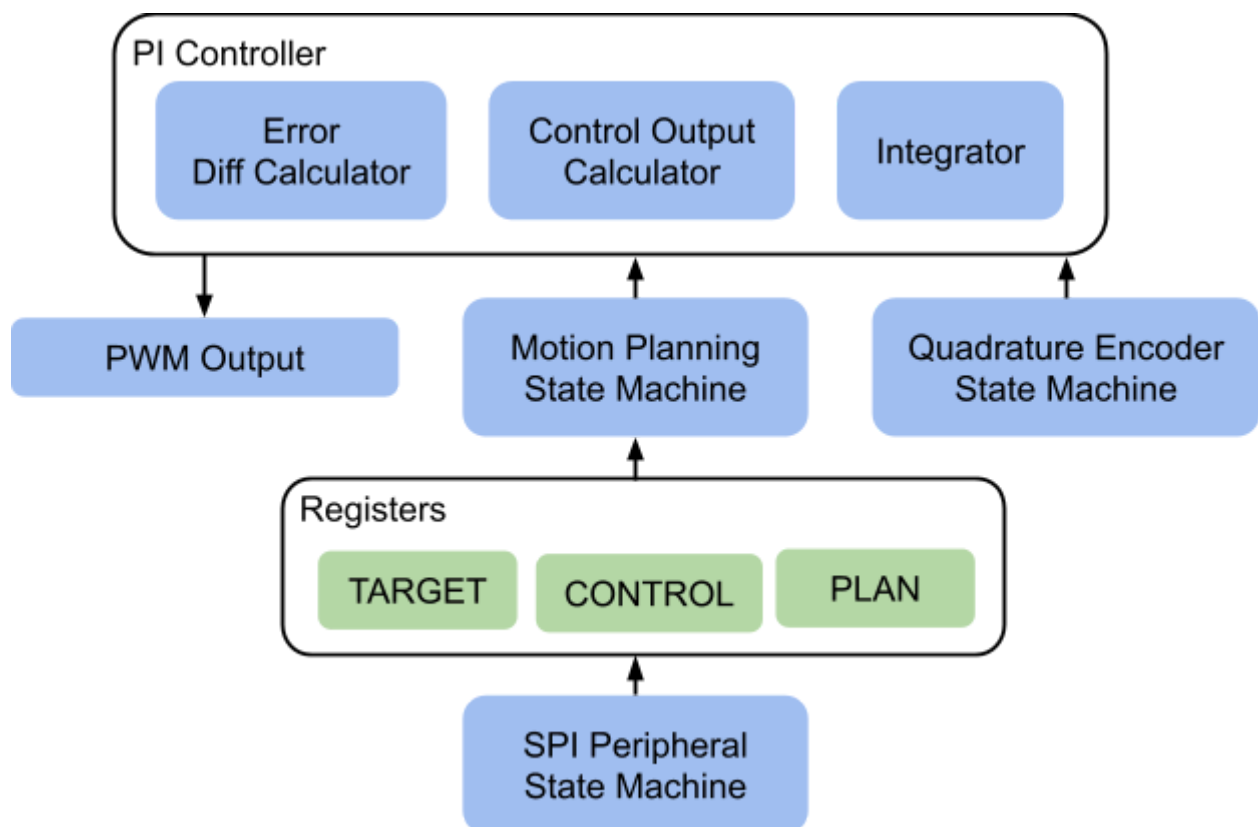


ECE298A Stepper Driver Chip

Purpose

A stepper motor driver ASIC that receives a desired motor position over SPI and then plans and executes a trajectory using trapezoidal motion profiling and closed loop PI control with feedback from an external quadrature encoder.

System Diagram



I/O Pin Assignment Table

Pin	Role
ui_in[0]	SPI clock
ui_in[1]	SPI chip select (active low)
ui_in[2]	SPI MOSI line
ui_in[3]	Encoder channel A (synchronized input)
ui_in[4]	Encoder channel B (synchronized input)
ui_in[5]	Motion enable (0 disables STEP output)
uo_out[0]	STEP (motor step pulses)
uo_out[1]	DIR (motor direction)
uo_out[2]	ENABLE (output motor enable)
uo_out[3]	Busy (trajectory active)
uo_out[4]	At target (settled within tolerance)
uo_out[5]	Fault (driver / encoder error)
clk	System clock
rst_n	Active-low reset

Proposed Specification

The following values are proposed design targets, to be confirmed through simulation, synthesis, and timing analysis.

Clock	
Maximum System Clock Frequency	10 MHz
SPI Peripheral	
Clock Polarity / CPOL	0

Clock Phase / CPHA	0
Maximum Clock Frequency	500 KHz
Bit Order	MSB first
Frame Schema	1 2 bit register address 13 bit payload* 0 3 bit status code *note that the upper bits for writes into smaller registers are automatically truncated.
Position & Motion	
Encoder Tick Storage	13 bit signed integer for target and current position
Control Loop Frequency	1 KHz
PI Coefficient Options	1, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$
Output	
PWM Width	8 bits
PWM Carrier Frequency	39 KHz (10 MHz / 256)
PWM Minimum Duty Cycle	1/256
PWM Maximum Duty Cycle	255/256

Registers

Name	Address	Width	Function
TARGET	0	13 bits	Target position
CONTROL	1	4 bits	The 2 MSB control kP, the 2 LSB control kI: 0b11 1 0b10 $\frac{1}{2}$ 0b01 $\frac{1}{4}$ 0b00 $\frac{1}{8}$
PLAN	2	6	Controls the maximum velocity and acceleration setpoints generated by the motion planner. The 3 MSB control the maximum acceleration: 0b000 500 steps/s ²

			0b111 100,000 steps/s ² The 3 LSB control the maximum velocity: 0b000 100 steps/s 0b111 10,000 steps/s
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Writes while busy or dropped writes are rejected.

The external stepper driver performs winding commutation and current limiting.

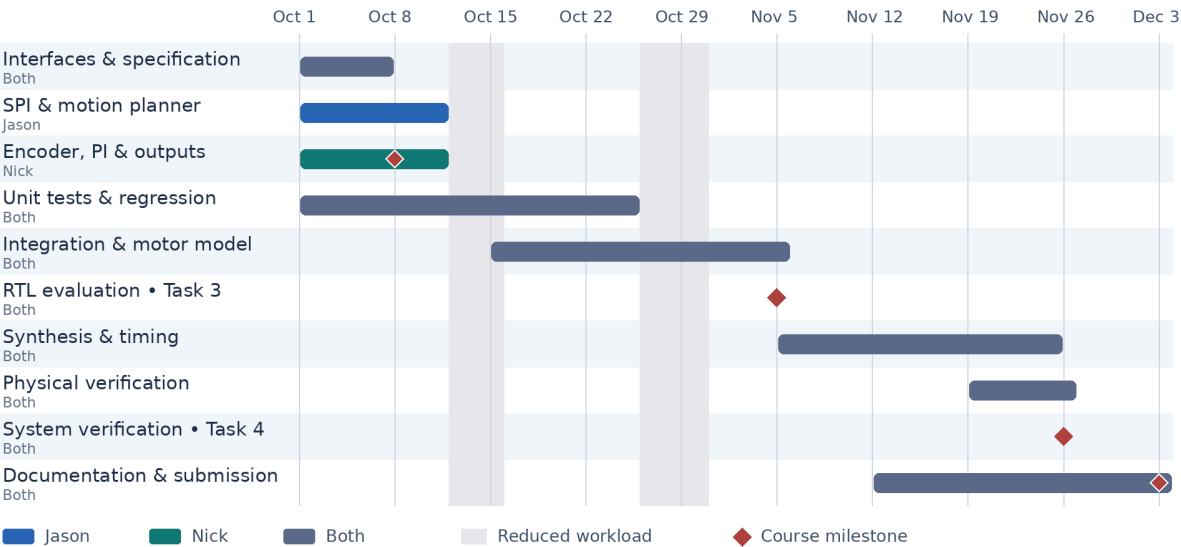
Verification: use Cocotb unit tests for SPI framing, quadrature direction, profile generation, arithmetic limits, PI saturation, pulse timing, and reset/fault behavior. Integrated tests use a discrete motor/encoder model for forward/reverse moves and disturbances. Completion requires passing GitHub Actions, synthesis and timing checks, and the selected Tiny Tapeout flow's physical verification.

Timeline for Completion

Planned work from October 1 to December 3, 2026. Reading week (Oct 12-15) and midterm week (Oct 26-30) are reserved mainly for catch-up and regression testing.

ECE298A • Project completion plan

October 1 – December 3, 2026



Milestones: Oct 8 • Task 2 Nov 5 • Task 3 Nov 26 • Task 4 Dec 3 • Final submission

Shaded periods: reading week (Oct 12-15) and midterm week (Oct 26-30).

Milestones: Oct 8 - Task 2, Verilog design and test; Nov 5 - Task 3, Verilog evaluation; Nov 26 - Task 4, system verification and submission readiness; Dec 3 - final submission and documentation. The course slide lists Task 1 on Sept 24, this proposal is being completed on Oct 1.

Who Does What

Proposed division of work:

Jason: SPI peripheral and register map, target-position interface, trapezoidal motion planner, unit tests and documentation for these modules.

Nick: quadrature decoder, PI controller and saturation/anti-windup, STEP/DIR generation unit tests and documentation for these modules.

Both: finalize interfaces and numerical scaling, integrate the design, review each other's code, build end-to-end Cocotb tests, run synthesis, timing and physical checks, resolve failures, prepare final documentation and Tiny Tapeout submission. Each owner is responsible for their module's implementation and tests, with the other member reviewing before integration.